FILTER DESIGN TEST CASE

TC-SAW (Temperature Compensated SAW) - Band 25 (1900 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P3 - Medium

1. OBJECTIVE

This document describes the design, simulation, and characterization of a TC-SAW (Temperature Compensated SAW) filter targeting Band 25 (1900 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on 36° YX $LiTaO_3$ substrate with a target center frequency of 1865.1 MHz and bandwidth of 220.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: TC-SAW (Temperature Compensated SAW)
- Frequency Band: Band 25 (1900 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: 36° YX $LiTaO_3$
- Package: Flip-Chip BGA
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Andover HQ Lab

This specification applies to all variants of the TC-SAW (Temperature Compensated SAW) design targeting Band 25 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1865.1 MHz
Bandwidth (3 dB):	220.7 MHz
Insertion Loss Target:	<= 1.3 dB
Return Loss Target:	>= 22.0 dB
Out-of-Band Rejection:	>= 46.0 dB
Isolation (Duplexer):	>= 43.0 dB
Q Factor:	1280
Temperature Coefficient:	-26.8 ppm/°C
Die Size:	1.1 x 1.6 mm
Wafer Size:	8-inch
Number of Resonators:	5
Electrode Material:	Mo

Passivation: SiO2

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the TC-SAW (Temperature Compensated SAW) template project.
2. Define the substrate stack: 36° YX $LiTaO_3$ with specified layer thicknesses.
3. Set design targets: center freq 1865.1 MHz, BW 220.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 36° YX $LiTaO_3$).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 5595 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.3 dB
- Return loss in passband shall be >= 22.0 dB
- Out-of-band rejection at +/- 331 MHz offset >= 46.0 dB
- Group delay variation across passband shall be <= 8 ns
- Temperature drift over -40°C to +85°C shall be <= 6.2 MHz
- Power handling shall be >= +35 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 88%

6. TEST RESULTS

Measurement Date: 2024-05-02
Devices Tested: 78
Insertion Loss (meas): 1.01 dB
Return Loss (meas): 23.7 dB
Rejection (meas): 49.3 dB
Isolation (meas): 48.1 dB
Q Factor (meas): 1280
Group Delay Variation: 1.0 ns
Power Handling: +26 dBm
Die Yield: 86.9%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

a) The TC-SAW (Temperature Compensated SAW) design demonstrated below-target performance for Band 25 (1900 MHz).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Temperature stability was within specification across full range.
e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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Approver: Dr. Hans Weber Date: 2024-05-06

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